

## LECTURE 11

## GMM I

## Definition

Suppose that an econometrician observes data  $\{W_i : i = 1, \dots, n\}$ , where  $W_i$  is a random  $p$ -vector. Let  $g$  be an  $l$ -dimensional function depending on  $W_i$  and the  $k$ -vector of parameters  $b$ :

$$g(W_i, b) = \begin{pmatrix} g_1(W_i, b) \\ \vdots \\ g_l(W_i, b) \end{pmatrix},$$

and  $g_j : \mathbb{R}^p \times \mathbb{R}^k \rightarrow \mathbb{R}$  for  $j = 1, \dots, l$ . The model is defined by the following *moment condition*:

$$\mathbb{E}[g(W_i, \beta)] = 0 \text{ for some } \beta \in \mathbb{R}^k. \quad (1)$$

Examples:

- **Linear regression.** Let  $W_i = (Y_i, X_i^\top)^\top$  and  $Y_i = X_i^\top \beta + U_i$ , where  $\beta \in \mathbb{R}^k$ , and  $\mathbb{E}[X_i U_i] = 0$ . In this case,  $g(W_i, b) = X_i (Y_i - X_i^\top b)$ ,  $l = k$ , and the moment condition is  $\mathbb{E}[X_i (Y_i - X_i^\top \beta)] = 0$ .
- **IV regression.** Let  $W_i = (Y_i, X_i^\top, Z_i^\top)^\top$ ,  $Y_i = X_i^\top \beta + U_i$ , where  $\beta \in \mathbb{R}^k$ , and  $\mathbb{E}[Z_i U_i] = 0$ , where  $Z_i$  is an  $l$ -vector. In this case,  $g(W_i, b) = Z_i (Y_i - X_i^\top b)$  with the moment condition  $\mathbb{E}[Z_i (Y_i - X_i^\top \beta)] = 0$ .
- **Lucas's model.** Suppose that in period  $t$ , investors derive utility from consumption  $C_t$ . Let  $R_{j,t}$  be the rate of return on risky asset  $j$ . Suppose that there are  $m$  assets. Assume that the utility function is  $\sum_{t=1}^{\infty} \delta^t C_t^{1-\alpha} / (1-\alpha)$ . In equilibrium, the returns on risky assets are determined by the following Euler equations:

$$\mathbb{E} \left[ \delta \left( \frac{C_{t+1}}{C_t} \right)^{-\alpha} (1 + R_{j,t+1}) \right] = 1, \quad j = 1, \dots, m.$$

Here  $W_t = (C_t, R_{1,t}, \dots, R_{m,t})$ ,  $b = (a, d)$ ,  $g_j(W_t, b) = d \left( \frac{C_{t+1}}{C_t} \right)^{-\alpha} (1 + R_{j,t+1}) - 1$  for  $j = 1, \dots, m$ , and the moment conditions are given by the above equations. Here,  $g$  is nonlinear in the parameters.

The model is *identified* if  $\mathbb{E}[g(W_i, \beta)] = 0$  and  $\mathbb{E}[g(W_i, \tilde{\beta})] = 0$  imply  $\beta = \tilde{\beta}$ , i.e., the solution of (1) is unique. The moment condition gives us  $l$  restrictions for  $k$  parameters. A necessary condition for the model to be identified is that  $l \geq k$ , i.e., there must be at least  $k$  restrictions. This necessary condition is called the *order condition*. The model is *underidentified* if the order condition fails.

When  $k = l$ , applying the MM principle, we can estimate  $\beta$  by the value of  $b$  that solves the sample analogue of (1):

$$n^{-1} \sum_{i=1}^n g(W_i, \hat{\beta}_n^{MM}) = 0.$$

When  $l > k$ , no  $b \in \mathbb{R}^k$  generally solves all  $l$  equations exactly. In this case, we can choose the value of  $b$  that makes the sample moments as close to zero as possible. Let  $A_n$  be a (possibly random)  $l \times l$  weight matrix such that  $A_n \rightarrow_p A$ , where  $A$  is non-random and has full rank  $l$ . The *Generalized Method of Moments (GMM) estimator* of  $\beta$  is defined to be the value of  $b$  that minimizes the weighted distance of  $n^{-1} \sum_{i=1}^n g(W_i, b)$  from zero:

$$\begin{aligned} \hat{\beta}_n^{GMM} &= \arg \min_{b \in B} \left\| A_n n^{-1} \sum_{i=1}^n g(W_i, b) \right\|^2 \\ &= \arg \min_{b \in B} \left( n^{-1} \sum_{i=1}^n g(W_i, b) \right)^\top A_n^\top A_n \left( n^{-1} \sum_{i=1}^n g(W_i, b) \right). \end{aligned} \quad (2)$$

The set  $B \subset \mathbb{R}^k$  is assumed to be compact. The matrix  $A^\top A$  is positive definite.

## Linear case

In this section, we discuss the IV regression example in detail. Here  $g$  is linear in the parameters. As in Lecture 10, we assume that some or all of the  $k$  regressors in  $X_i$  are endogenous:

$$\mathbb{E}[X_i U_i] \neq 0,$$

and that the  $l$  instruments  $Z_i$  are weakly exogenous:

$$\mathbb{E}[Z_i U_i] = 0.$$

The model is identified if the following *rank condition* is satisfied:

$$\text{rank}(\mathbb{E}[Z_i X_i^\top]) = k.$$

If the rank condition is satisfied and  $l = k$ , we say that the model is *exactly* or *just* identified. We say that the model is *overidentified* if the rank condition is satisfied and  $l > k$  (there are more instruments than parameters). Unlike in Lecture 10, the model may be overidentified here.

In the linear IV regression case,  $\hat{\beta}_n^{GMM}$  is the minimizer of

$$\left( n^{-1} \sum_{i=1}^n Z_i (Y_i - X_i^\top b) \right)^\top A_n^\top A_n \left( n^{-1} \sum_{i=1}^n Z_i (Y_i - X_i^\top b) \right),$$

and is given by the following expression:

$$\hat{\beta}_n^{GMM} = \left( \sum_{i=1}^n X_i Z_i^\top (A_n^\top A_n) \sum_{i=1}^n Z_i X_i^\top \right)^{-1} \sum_{i=1}^n X_i Z_i^\top (A_n^\top A_n) \sum_{i=1}^n Z_i Y_i.$$

We show next that the GMM estimator is consistent. We need the following assumptions.

- $\{(Y_i, X_i, Z_i) : i = 1, \dots, n\}$  are iid.
- $Y_i = X_i^\top \beta + U_i$ , where  $\beta \in \mathbb{R}^k$ .
- $\mathbb{E}[Z_i U_i] = 0$ .
- $\mathbb{E}[Z_i X_i^\top]$  has rank  $k$ .
- $A_n \rightarrow_p A$ , where  $A$  has rank  $l \geq k$ .
- $\mathbb{E}[X_{i,j}^2] < \infty$  for all  $j = 1, \dots, k$ .
- $\mathbb{E}[Z_{i,j}^2] < \infty$  for all  $j = 1, \dots, l$ .

Write

$$\hat{\beta}_n^{GMM} = \beta + \left( n^{-1} \sum_{i=1}^n X_i Z_i^\top (A_n^\top A_n) n^{-1} \sum_{i=1}^n Z_i X_i^\top \right)^{-1} n^{-1} \sum_{i=1}^n X_i Z_i^\top (A_n^\top A_n) n^{-1} \sum_{i=1}^n Z_i U_i.$$

The last two assumptions imply that

$$\mathbb{E}|X_{i,r} Z_{i,s}| < \infty \quad \text{for all } r = 1, \dots, k \text{ and } s = 1, \dots, l.$$

By the WLLN,

$$n^{-1} \sum_{i=1}^n X_i Z_i^\top \rightarrow_p E[X_i Z_i^\top].$$

Since  $A_n \rightarrow_p A$ , we also have that

$$n^{-1} \sum_{i=1}^n X_i Z_i^\top (A_n^\top A_n) n^{-1} \sum_{i=1}^n Z_i X_i^\top \rightarrow_p E[X_i Z_i^\top] (A^\top A) E[Z_i X_i^\top].$$

Further, since  $E[Z_i X_i^\top]$  has rank  $k$  and  $A$  has rank  $l \geq k$ , it follows that the  $k \times k$  matrix  $E[X_i Z_i^\top] (A^\top A) E[Z_i X_i^\top]$  has full rank  $k$  and is therefore invertible. Consequently, by Slutsky's Theorem,

$$\left( n^{-1} \sum_{i=1}^n X_i Z_i^\top (A_n^\top A_n) n^{-1} \sum_{i=1}^n Z_i X_i^\top \right)^{-1} \rightarrow_p (E[X_i Z_i^\top] (A^\top A) E[Z_i X_i^\top])^{-1}.$$

Next, by the WLLN,

$$n^{-1} \sum_{i=1}^n Z_i U_i \rightarrow_p 0,$$

and thus  $\hat{\beta}_n^{GMM} \rightarrow_p \beta$ .

To show asymptotic normality, we will need the following three assumptions in addition to the above.

- $E[Z_{i,j}^4] < \infty$  for all  $j = 1, \dots, l$ .
- $E[U_i^4] < \infty$ .
- $E[U_i^2 Z_i Z_i^\top]$  is positive definite.

Write

$$n^{1/2} (\hat{\beta}_n^{GMM} - \beta) = \left( n^{-1} \sum_{i=1}^n X_i Z_i^\top (A_n^\top A_n) n^{-1} \sum_{i=1}^n Z_i X_i^\top \right)^{-1} n^{-1} \sum_{i=1}^n X_i Z_i^\top (A_n^\top A_n) n^{-1/2} \sum_{i=1}^n Z_i U_i.$$

The last two assumptions imply that the variance of  $Z_i U_i$ ,  $E[U_i^2 Z_i Z_i^\top]$ , is finite. By the CLT,

$$n^{-1/2} \sum_{i=1}^n Z_i U_i \rightarrow_d N(0, E[U_i^2 Z_i Z_i^\top]).$$

Define

$$\begin{aligned} Q &= E[Z_i X_i^\top], \\ \Omega &= E[U_i^2 Z_i Z_i^\top]. \end{aligned}$$

Combining the above results, we have

$$n^{1/2} (\hat{\beta}_n^{GMM} - \beta) \rightarrow_d N(0, V),$$

where  $V$  takes the sandwich form:

$$V = (Q^\top A^\top A Q)^{-1} Q^\top A^\top A \Omega A^\top A Q (Q^\top A^\top A Q)^{-1}.$$

The variance-covariance matrix  $V$  can be estimated by replacing  $A$ ,  $Q$ , and  $\Omega$  with their consistent estimators  $A_n$ ,  $\hat{Q}_n$ , and  $\hat{\Omega}_n$ , respectively, where

$$\begin{aligned} \hat{Q}_n &= n^{-1} \sum_{i=1}^n Z_i X_i^\top, \\ \hat{\Omega}_n &= n^{-1} \sum_{i=1}^n \hat{U}_i^2 Z_i Z_i^\top, \end{aligned}$$

and  $\hat{U}_i = Y_i - X_i^\top \hat{\beta}_n^{GMM}$ .

## General case

In the general case, the GMM estimator minimizes the nonlinear function in (2). Usually, we do not have a closed-form expression for  $\hat{\beta}_n^{GMM}$ , and the minimization must be done using numerical procedures. Nevertheless, under general regularity conditions, it is possible to show that  $\hat{\beta}_n^{GMM}$  is consistent and asymptotically normal. We provide heuristic arguments for consistency and asymptotic normality.

Since the criterion function in (2) involves averages, we should expect that

$$\left\| A_n n^{-1} \sum_{i=1}^n g(W_i, b) \right\|^2 \rightarrow_p \|A E[g(W_i, b)]\|^2. \quad (3)$$

Assuming that the model is uniquely identified,  $E[g(W_i, b)] = 0$  if and only if  $b = \beta$ . Since  $\|A E[g(W_i, b)]\|^2 > 0$  for all  $b \neq \beta$ , the true value  $\beta$  is the unique minimizer of  $\|A E[g(W_i, b)]\|^2$ . Intuitively,  $\hat{\beta}_n^{GMM}$  is consistent because

$$\begin{aligned} \hat{\beta}_n^{GMM} &= \arg \min_{b \in B} \left\| A_n n^{-1} \sum_{i=1}^n g(W_i, b) \right\|^2 \\ &\rightarrow_p \arg \min_{b \in B} \|A E[g(W_i, b)]\|^2 \\ &= \beta. \end{aligned}$$

The formal proof of consistency requires a number of regularity conditions, such as *uniform* convergence over  $b \in B$  in (3), compactness of  $B$ , and  $\beta$  being an interior point of  $B$ .

For asymptotic normality,  $\hat{\beta}_n^{GMM}$  solves the first-order conditions:

$$\left( n^{-1} \sum_{i=1}^n \frac{\partial g(W_i, \hat{\beta}_n^{GMM})}{\partial b^\top} \right)^\top A_n^\top A_n n^{-1} \sum_{i=1}^n g(W_i, \hat{\beta}_n^{GMM}) = 0. \quad (4)$$

(In fact, it is sufficient if  $\hat{\beta}_n^{GMM}$  solves the first-order conditions approximately, i.e., the right-hand side of the above equation may be  $o_p(n^{-1/2})$  rather than zero.) Next, using the expansion of  $g(W_i, \hat{\beta}_n^{GMM})$  around  $g(W_i, \beta)$  (the element-by-element mean value theorem), we obtain

$$g(W_i, \hat{\beta}_n^{GMM}) = g(W_i, \beta) + \frac{\partial g(W_i, \hat{\beta}_n^*)}{\partial b^\top} (\hat{\beta}_n^{GMM} - \beta), \quad (5)$$

where  $\hat{\beta}_n^*$  is between  $\hat{\beta}_n^{GMM}$  and  $\beta$ . Substitution of (5) into (4) gives

$$\begin{aligned} 0 &= \left( n^{-1} \sum_{i=1}^n \frac{\partial g(W_i, \hat{\beta}_n^{GMM})}{\partial b^\top} \right)^\top A_n^\top A_n n^{-1} \sum_{i=1}^n g(W_i, \beta) \\ &\quad + \left( n^{-1} \sum_{i=1}^n \frac{\partial g(W_i, \hat{\beta}_n^{GMM})}{\partial b^\top} \right)^\top A_n^\top A_n \left( n^{-1} \sum_{i=1}^n \frac{\partial g(W_i, \hat{\beta}_n^*)}{\partial b^\top} \right) (\hat{\beta}_n^{GMM} - \beta). \end{aligned}$$

We can write

$$\begin{aligned}
& n^{1/2} \left( \widehat{\beta}_n^{GMM} - \beta \right) \\
&= - \left( \left( n^{-1} \sum_{i=1}^n \frac{\partial g(W_i, \widehat{\beta}_n^{GMM})}{\partial b^\top} \right)^\top A_n^\top A_n \left( n^{-1} \sum_{i=1}^n \frac{\partial g(W_i, \widehat{\beta}_n^*)}{\partial b^\top} \right) \right)^{-1} \\
&\quad \times \left( n^{-1} \sum_{i=1}^n \frac{\partial g(W_i, \widehat{\beta}_n^{GMM})}{\partial b^\top} \right)^\top A_n^\top A_n n^{-1/2} \sum_{i=1}^n g(W_i, \beta).
\end{aligned}$$

Since  $E[g(W_i, \beta)] = 0$ , under regularity conditions,

$$n^{-1/2} \sum_{i=1}^n g(W_i, \beta) \rightarrow_d N \left( 0, E[g(W_i, \beta) g(W_i, \beta)^\top] \right).$$

(The asymptotic variance depends on the unknown  $\beta$ .) Since  $\widehat{\beta}_n^{GMM}$  is consistent, and, as a result,  $\widehat{\beta}_n^* \rightarrow_p \beta$  as well, under regularity conditions,

$$\begin{aligned}
n^{-1} \sum_{i=1}^n \frac{\partial g(W_i, \widehat{\beta}_n^{GMM})}{\partial b^\top} &\rightarrow_p E \left[ \frac{\partial g(W_i, \beta)}{\partial b^\top} \right], \\
n^{-1} \sum_{i=1}^n \frac{\partial g(W_i, \widehat{\beta}_n^*)}{\partial b^\top} &\rightarrow_p E \left[ \frac{\partial g(W_i, \beta)}{\partial b^\top} \right],
\end{aligned}$$

and that the matrix

$$\left( E \left[ \frac{\partial g(W_i, \beta)}{\partial b^\top} \right] \right)^\top A^\top A \left( E \left[ \frac{\partial g(W_i, \beta)}{\partial b^\top} \right] \right)$$

is invertible. Then,

$$n^{1/2} \left( \widehat{\beta}_n^{GMM} - \beta \right) \rightarrow_d N(0, V),$$

where

$$\begin{aligned}
V &= (Q^\top A^\top A Q)^{-1} Q^\top A^\top A \Omega A^\top A Q (Q^\top A^\top A Q)^{-1}, \\
Q &= E \left[ \frac{\partial g(W_i, \beta)}{\partial b^\top} \right], \\
\Omega &= E[g(W_i, \beta) g(W_i, \beta)^\top].
\end{aligned}$$

The variance-covariance matrix  $V$  can be estimated by replacing  $A$ ,  $Q$ , and  $\Omega$  with their consistent estimators  $A_n$ ,  $\widehat{Q}_n$ , and  $\widehat{\Omega}_n$ , respectively, where

$$\begin{aligned}
\widehat{Q}_n &= n^{-1} \sum_{i=1}^n \frac{\partial g(W_i, \widehat{\beta}_n^{GMM})}{\partial b^\top}, \\
\widehat{\Omega}_n &= n^{-1} \sum_{i=1}^n g(W_i, \widehat{\beta}_n^{GMM}) g(W_i, \widehat{\beta}_n^{GMM})^\top.
\end{aligned}$$